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Association between accounting performance measures and stock prices

A test of the life cycle hypothesis*

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This paper posits that stock market response to two accounting performance measures – sales growth and capital investment – is a function of firm life cycle stage. Firms are grouped into various life cycle portfolios using dividend payout, sales growth, and age. As predicted, the empirical results indicate a monotonic decline in the response coefficients of unexpected sales growth and unexpected capital investment from the growth to the stagnant stages. Additional analysis suggests that this relation is not driven by a firm size effect, risk differences, or measurement error in the proxies for performance measures.

1. Introduction

We investigate the implications of a life cycle theory of the firm for the stock market response to accounting performance measures. A sizeable literature (detailed below) suggests that (1) changes in sales growth and capital expenditure signal the strategic emphasis of the firm (e.g., capturing market share and increasing capital capacity versus cost trimming) and (2) the cost effectiveness of

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a strategy is a function of life cycle stage. We predict and find empirically that stock market reactions to sales growth and capital expenditure are functions of life cycle stage.

Our argument is motivated by strategic prescriptions prevalent in the management, marketing, and economics literature, which originated with the Boston Consulting Group (1968) (hereafter BCG). BCG's underlying idea is that a firm maximizes revenue growth early in its life cycle, to create permanent cost or demand advantages over competitors, but in its mature stage market growth slows and investments are less rewarding [Porter (1980, p. 248)]. Economists rationalize this model in a variety of ways [Spence (1977, 1979, 1981), Karnani (1984), and Wernerfelt (1985)]. For example, Spence rationalizes early growth maximization through irreversibility of investment with a growth constraint, and early capacity maximization as a deterrent to potential entrants.

This reasoning suggests that performance measures differ across life cycle stages, which is evident in the management accounting literature [Richardson and Gordon (1980), Rappaport (1981), and National Association of Accountants (NAA) (1986)]. For example, NAA argues that '[a]t each stage of growth in an entity's life cycle, different measures of financial performance take on varying degrees of importance. Therefore, neither growth nor net income nor cash flows nor return on investment should be emphasized to the exclusion of other meaningful measures' (p. 13).

We use a market-based approach to test if stock market reactions to sales growth and capital expenditure are functions of life cycle stage. Specifically, we hypothesize that unexpected positive sales growth and capital expenditure are most (least) valued by the capital market during the firm's growth (stagnant) stage. We choose dividend payout, sales growth, and firm age as indicators of life cycle stage. We classify firm-years into life cycle groups, first using individual variables and then using a composite score obtained from all variables (univariate and multivariate groupings).

We test our hypotheses by regressing cumulative abnormal returns (CAR) on differenced earnings, capital expenditure, and sales growth variables, with slope dummies for various life cycle groups formed by the univariate and multivariate approaches. The results are consistent with our predictions. In the univariate groupings, the response coefficients of unexpected sales growth and unexpected capital expenditure are higher for low dividend payout (young/high sales growth) compared to high dividend payout (old/low sales growth) firms. The multivariate results indicate a monotonic decline both in the magnitude and statistical significance of the response coefficients from the growth to the stagnant groups. Directional t-tests of differential response coefficients also support our predictions. Taken together, the evidence suggests that the stock market reaction to performance measures is a function of life cycle stage.

Additional analyses suggest that differential response coefficients are not driven by a firm size effect or risk differences, and that differences in time-series

properties of the performance measures do not systematically contribute to measurement error across life cycle groups. Overall, our results highlight the role of nonearnings data in explaining stock returns and suggest that strategic issues are important in performance evaluation.

The remainder of the paper is organized as follows. Section 2 develops the hypotheses. The sample selection procedure, life cycle descriptors, definition of the independent and dependent variables, and regression model are provided in section 3. Empirical results and alternative explanations are discussed in sections 4 and 5, respectively. Limitations and concluding remarks are provided in the final section.

2. Hypothesis development

The central idea in business strategy is to create a 'lasting' cost or demand advantage over competitors [BCG (1968)]. Cost advantages might include building capacity to achieve economies of scale, while demand advantages stem from building a large market share, either of which creates a barrier to entry. Life cycle theory suggests that appropriate growth and capital capacity strategies depend on the firm's life cycle stage. Early growth maximization is suggested in many business strategy texts [e.g., Porter (1980) and BCG (1968)]. Similarly, a firm can strategically position itself by creating capital capacity early in its life cycle. Thus, the benefit–cost ratio of acquiring market share and building capacity is highest in the firm's early life cycle stages.

Economists attempt to derive these prescriptions analytically. Spence (1977) shows that firms can deter entry by creating capacity and incurring significant capital expenditures early in the life cycle, making the product market unattractive to potential entrants. Spence (1979) further rationalizes early growth maximization through irreversibility of investments combined with a growth constraint. Spence shows that the individual firm invests rapidly until the present value of the net marginal profitability of capital is zero. Firms that can grow rapidly can make preemptive investments.

Demand- and supply-side learning are considered important by strategists in many industries. Spence (1981) argues that the learning curve creates entry barriers and protection from competition by creating cost advantages for those who achieve large market shares early in the life cycle. Even though cost advantages are only temporary, they significantly impact market share and profitability. Wernerfelt (1985) shows that in the presence of learning curves, declining price sensitivity, and declining growth rates, growth maximization early in the life cycle can be a means of profit maximization.

Since acquisition of market share and capital capacity are highly valued in early life cycle stages, it is reasonable to expect a higher stock price reaction to unexpected sales growth (or unexpected capital expenditure) in the early life

cycle stages. This leads to the following research hypotheses (stated in alternative form):

(1) Unexpected positive sales growth is most (least) highly valued by the capital market during the growth (stagnant) stage.

(2) Unexpected positive capital expenditure is most (least) highly valued by the capital market during the growth (stagnant) stage.

Even though our hypotheses relate only to unexpected capital expenditures and sales growth, we also investigate the response to unexpected earnings, to mitigate omitted variables concerns. The association between earnings and stock prices is well documented in the accounting literature [e.g., Ball and Brown (1968) and Beaver (1968)]. Further, profitability measures are frequently referenced in the business strategy and management accounting literatures in the context of life cycle analysis [see Porter (1980) and NAA (1986)]. Unlike market share and capital capacity, there are no analytical models offering a directional relation between unexpected earnings and life cycle stage. Lacking adequate theoretical arguments, we develop no specific life cycle hypothesis with respect to unexpected earnings [see Rao (1989) for an investigation of the informativeness of earnings changes as a function of firm age].

3. Research design

3.1. Sample selection

Our initial sample includes 3,686 firms found on both the 1987 Compustat (Annual and Research) and CRSP tapes. We eliminate 740 utilities, insurance, and financial institutions. We exclude 1,037 firms not on the CRSP or Compustat tape for at least six years (explained in more detail below). We also eliminate 84 firms lacking sufficient data to estimate the market model parameters, leaving a potential sample of 1,825 firms with 14,258 firm-years of data.

We further restrict our analysis to those years for which life cycle data are available for at least 1,000 firms. This requirement is satisfied for the years 1976 to 1986. The year 1987 is excluded since our version of Compustat lacks necessary updated information. Additional restrictions are imposed by data requirements of the individual tests performed. Elimination due to specific missing data items (noted where appropriate) and deletion of outliers in the regression analyses cause sample size to vary from 11,768 to 13,882 firm-years across the various tests.[1]

[1]To mitigate the impact of outliers, we delet firm-years having extreme values ($\pm 300\%$) in independent variables.

Table 1

Expectations for firm-specific descriptors of life cycle stages.[a]

	Life cycle descriptors			
Life cycle stages	*DP*	*SG*	*CEV*	*AGE*
Growth	Low	High	High	Young
Mature	Medium	Medium	Medium	Adult
Stagnant	High	Low	Low	Old

[a]*DP*, *SG*, *CEV*, and *AGE* refer to dividend payout, sales growth, capital expenditure divided by value of the firm (market value of equity plus book value of long-term debt), and firm age, respectively.

3.2. *Life cycle descriptors*

We classify firms into life cycle stages using both univariate and multivariate ranking procedures. We use four classification variables: (1) annual dividend as a percentage of income (*DP*), (2) percent sales growth (*SG*), (3) capital expenditure as a percentage of total value of the firm (*CEV*), and (4) age of the firm (*AGE*).[2] We choose variables based on their frequent reference in the economics, management, and management accounting literature in similar contexts [e.g., Spence (1979, p. 1), Kotler (1980, table 12-1, p. 301), and NAA (1986, table 1, p. 10)]. Expectations regarding these life cycle descriptors are provided in table 1. The intuition is straightforward. Firms in early life cycle stages, on average, exhibit higher sales growth. Growth firms invest larger proportional amounts in plant and equipment, and have lower dividend payout ratios given their opportunity set of positive net present value projects. Younger firms are more likely to have new products.

We compute firm-specific financial variables to identify the life cycle stage from Compustat data as follows:

$$DP_t = (DIV_t/IBED_t) \times 100, \tag{1}$$

$$SG_t = ((SALES_t - SALES_{t-1})/SALES_{t-1}) \times 100, \tag{2}$$

$$CEV_t = (CE_t/VALUE_t) \times 100, \tag{3}$$

[2]We thank the referee for suggesting firm age as a variable.

where (number in parentheses refers to the Compustat data item)

DIV_t = common dividends in year t (21),
$IBED_t$ = income before extraordinary items and discontinued operations in year t (18),
$SALES_t$ = net sales in year t (12),
CE_t = capital expenditure in year t (128), and
$VALUE_t$ = market value of equity plus book value of long-term debt at the end of year t.

The financial variables chosen as life cycle descriptors are directly related to firm risk, so firms sorted on these variables could have a differential response to performance measures, even without life cycle considerations. For example, prior research [Easton and Zmijewski (1989) and Collins and Kothari (1989)] demonstrates a relation between earnings response coefficients and risk, and a correlation between sales and earnings. To minimize the effect of possible correlation of risk with life cycle stage, we choose firm age as a nonfinancial life cycle descriptor, and we control for risk in our specification tests.

The three financial life cycle descriptors are calculated for each year for each sample firm. Then, for each firm-year, median values of the variables (denoted *MDP*, *MSG*, and *MCEV*) are computed using the prior five years' data. This requires at least six years of data availability for each firm. Information on the *AGE* variable is obtained from Moody's manuals. For each firm, we obtain the year in which the business was originally formed (denoted *BYEAR*).[3] *AGE* is computed as the difference between the current year and *BYEAR* for each firm-year.

We rank firms on each of the four life cycle descriptors (*MDP*, *MSG*, *MCEV*, and *AGE*)[4] and group them into various life cycle stages [Low, Medium, and High (Young, Adult, and Old for the *AGE* variable)] based on table 1. This approach is employed separately in each year to allow for temporal shifts in the life cycle stage of sample firms.

Once a firm-year is assigned to a group, it is given a score (growth = 1, mature = 2, and stagnant = 3). For example, a firm-year with a low *SG* (a candidate for the 'stagnant' stage) is given a score of three for the sales growth

[3]If the date of business formation is not available, then the year in which the firm was originally *incorporated* is considered the *BYEAR*. If two firms are merged, then for each firm, *BYEAR* is obtained, and the earlier of the two *BYEAR*'s is considered the year of business formation for the new corporation. Missing information on *BYEAR* reduces the sample to 1,777 firms (13,466 firm years).

[4]The choice of median value for dividend payout reduces the problem of negative values for *MDP*. Only 68 firm-years are associated with negative *MDP*'s. These are assigned to the 'High *MDP*' group. Tables 5 through 8 were replicated using median dividend yield (*MDY*), instead of *MDP*, as a life cycle descriptor. The results (not reported here) are qualitatively similar.

variable and a firm-year with a low *DP* (a candidate for the 'growth' stage) is given a score of one for the dividend payout variable.

In our multivariate analysis, we create five life cycle groups using *MDP*, *MSG*, and *AGE* as descriptors, based on a composite score obtained by summing the individual variable scores.[5] Although based on a simple summation, the composite score incorporates some interactions among the individual variables. For example, a low dividend payout could signal either high growth opportunities or cash flow problems. While a stagnant firm with cash flow problems would exhibit a low dividend payout, it is unlikely to rank high on the remaining variables, and thus should end up properly classified as a stagnant firm.[6]

When we use the composite scoring technique, 73.35% of firms remain in the same life cycle group compared to the prior year. The percentage decreases to 53.29% (44.09%) when comparing to three years (five years) before. A significant portion of this variability is due to shifts to adjacent groups. Compared with one (three/five) year(s) prior to the current year, 96.46% (88.77%/84.06%) of firms remain within one adjacent group.

3.3. *Definition of independent and dependent variables*

Three financial performance measures are used as independent variables:

(1) unexpected earnings $[\Delta IBED = (IBED_t - IBED_{t-1})/MVE_{t-1}]$,

(2) unexpected capital expenditures $[\Delta CE = (CE_t - CE_{t-1})/MVE_{t-1}]$,

(3) unexpected sales growth $[\Delta SG = SG_t - SG_{t-1}]$,[7]

where MVE_{t-1} is market value of equity at the end of time $t-1$, and other variables are as already defined.

The dependent variable is the cumulative abnormal return (*CAR*) from the market model. The market model parameters are derived from OLS regression using returns from month -1 to month -60 (the estimation period), where month zero is the first month of the fiscal year. *CAR* is computed by cumulating

[5] *MCEV* is eliminated as a descriptor due to its low explanatory power. See sections 4.2 and 4.3 for additional detail.

[6] As the referee points out, future research which explicitly considers interaction terms for the life cycle measures or an alternative weighting scheme may provide additional insights. However, at present, we have no means of further weighting the interactions, other than an ad hoc approach.

[7] We adopt random walk expectation models for sales growth and capital expenditure. Future research might include financial analysts' forecasts (e.g., Value Line) as expectations proxies. We credit the referee for this suggestion.

Unexpected earnings and unexpected capital expenditure are scaled by *MVE* to mitigate the impact of cross-sectional differences in firm size. Scaling is not needed for unexpected sales growth, since this variable is based on a pure number.

abnormal returns from the fourth month of the relevant fiscal year to the third month following the end of the fiscal year.[8]

3.4. Regression model

We estimate the following regression model to test the life cycle hypotheses:

$$CAR = \sum_i D_i[\alpha_{0i} + \alpha_{1i}\Delta IBED + \alpha_{2i}\Delta CE + \alpha_{3i}\Delta SG] + \varepsilon. \tag{4}$$

D_i is a dummy variable which takes a value of zero or one and the summation is over the n life cycle categories (either three or five). In the univariate procedure, D_1 equals one when a firm-year is assigned to Low *MDP* (High *MSG*/High *MCEV*/Young) group, D_2 equals one when a firm-year is assigned to Medium *MDP* (Medium *MSG*/Medium *MCEV*/Adult) group, and D_3 equals one when a firm-year is assigned to High *MDP* (Low *MSG*/Low *MCEV*/Old) group. The life cycle hypotheses translate into restrictions on α_2 and α_3 across subsamples. For example, if the dummy variables correspond to levels of *MDP*, then we expect α_2 and α_3 of the Low *MDP* group to be greater than those of the High *MDP* group. The following statistical hypotheses (stated in alternative form) are tested:

$$H_1: \quad \alpha_{j1} - \alpha_{j2} \geq 0, \quad j = 2,3,$$

$$H_2: \quad \alpha_{j1} - \alpha_{j3} \geq 0, \quad j = 2,3,$$

$$H_3: \quad \alpha_{j2} - \alpha_{j3} \geq 0, \quad j = 2,3$$

In the multivariate procedure, we assign firm-years to five life cycle groups [Growth (G), Growth/Mature (GM), Mature (M), Mature/Stagnant (MS), and Stagnant (S)] based on the composite score.[9] We expect α_2 and α_3 to monotonically decline from the Growth to Stagnant portfolios. In this procedure, D_1 takes a value of one when a firm-year belongs to the Growth group, whereas D_5 takes a value of one when a firm-year belongs to the Stagnant group. D_2 through

[8]The cumulation period is selected to cover the period between consecutive annual report release dates. Firms with as few as five data points in the estimation period are included. Of the 14,258 betas estimated, sixteen are estimated with less than twenty observations, and 14,066 betas are estimated with no missing observations.

[9]The range of the composite score depends on the number of grouping variables. For example, with three variables, the composite score ranges from three to nine. Firm-years are assigned to the five life cycle groups to yield an approximately equal number of observations in each group.

Table 2

Summary descriptive statistics.[a]

Variable[b]	Mean	Std. dev.	Quartiles		
			0.25	0.50	0.75
MROA	6.004	4.262	3.501	5.771	8.306
MDP	22.628	19.598	0.439	22.598	34.991
MSG	11.878	10.751	6.138	11.400	16.990
MCEV	10.373	6.785	5.455	8.950	13.618
MVALUE	663.440	2495.820	30.337	106.408	433.953
β	1.261	0.900	0.918	1.206	1.535
AGE	56.922	31.466	31.000	53.000	77.000

[a]Based on 13,882 firm-years, except for the *AGE* variable, where only 13,686 firm-years of data are available. All median variables are recalculated for each firm-year using the prior five years' data.

[b]*MROA* = median of income before extraordinary items and discontinued operations divided by market value of equity, expressed as a percentage;
MDP = median of common dividends divided by income before extraordinary items and discontinued operations, expressed as a percentage;
MSG = median annual percentage sales growth;
MCEV = median of capital expenditure divided by the market value of equity plus book value of long-term debt, expressed as a percentage;
MVALUE = median market value of equity plus book value of debt;
β = estimate of market model slope coefficient, using prior five years' data;
AGE = age of the firm, computed as the difference between the current year and the year in which the business was originally formed.

D_4 are dummy variables representing the middle three groups. We test the following statistical hypotheses (stated in alternative form):

$$H_4: \quad \alpha_{j1} - \alpha_{j3} \geq 0, \quad j = 2,3,$$

$$H_5: \quad \alpha_{j1} - \alpha_{j5} \geq 0, \quad j = 2,3,$$

$$H_6: \quad \alpha_{j3} - \alpha_{j5} \geq 0, \quad j = 2,3.$$

Statistical testing is restricted to the Growth, Mature, and Stagnant groups to minimize the number of statistical comparisons.

4. Analysis of the results

4.1. Descriptive information

Table 2 provides descriptive information on the life cycle descriptors (*MDP*, *MSG*, *MCEV*, and *AGE*), median firm value (*MVALUE*), beta (β) of the sample

firm-years, and median return on assets (*MROA*).[10] The mean of *MVALUE* is driven by extreme outliers. Higher mean and median values for beta (i.e., greater than 1.00) are due to inclusion of many smaller AMEX firms and exclusion of utilities from the sample.

Table 3 provides median values for the same variables grouped by levels of the life cycle descriptors. For example, panel A of table 3 provides median values for three groups formed by levels of MDP. Therefore, in each panel, the monotonic increase in median values reported for one variable is by construction. Table 3 indicates the interrelations among the various performance measures and other firm characteristics. In general, observed relations among *MDP*, *MSG*, and *AGE* are consistent with our expectations, supporting their choice as life cycle descriptors. Specifically, there is a positive association between dividend payout and firm age, and a negative association between sales growth and firm age.

4.2. Industry-level life cycles information

We next investigate the reasonableness of our life cycle descriptors at the industry level. We use data on all firm-years to obtain industry median values (based on four-digit SIC codes) for each life cycle descriptor (denoted *IMDP*, *IMSG*, *IMCEV*, and *IMAGE*). We exclude industries with less than 40 firm-years of data. We rank industries and then consider the reasonableness of the classifications for the top and bottom 20 industries. The evidence (not presented) suggests that dividend payout, sales growth, and firm age are reasonable proxies for life cycle stage. The low dividend payout, high sales growth, and young industries include those generally considered 'growth' industries during the period studied (e.g., computers and semiconductors in the low dividend payout category; computers and hospitals in the high sales growth category; and semiconductors and consulting in the new industries category). However, one can point out 'inappropriate' classifications obtained using the individual life cycle descriptors.

Capital expenditure provides little insight on life cycle stage. There is a concentration of capital-intensive firms in the 'High *IMCEV*' category, many of which are not typically considered 'growth' industries. Inter-industry differences in *IMCEV* appear more related to an industry's production function than to life cycle stage. Thus, *IMCEV*, in the univariate industry level analysis, proxies poorly for life cycle stage.

Combining the three proxies (*IMDP*, *IMSG*, and *IMAGE*) should reduce the probability of misclassification. To test this argument, we first rank industries

[10]Return on assets is defined as *IBED* divided by total assets. Consistent with the presentation for the life cycle descriptors, we provide descriptive information on *median* values for total firm value and return on assets.

Table 3

Median values for alternative life cycle groups.[a,b]

Group	*n*	*MROA*	*MDP*	*MSG*	*MCEV*	*MVALUE*	*β*	*AGE*
Panel A: Median values for groups formed by levels of dividend payout (MDP)								
Low	4,292	3.332	0.000	10.140	9.182	36.474	1.471	34.0
Medium	4,668	6.637	21.625	13.640	9.157	124.640	1.246	50.0
High	4,922	6.521	39.535	10.201	8.615	269.163	1.019	72.0
Panel B: Median values for groups formed by levels of sales growth (MSG)								
Low	4,825	4.179	21.549	3.890	8.887	58.971	1.178	55.0
Medium	4,804	6.322	26.943	11.970	8.900	149.681	1.151	60.0
High	4,253	7.118	18.602	20.430	9.109	143.588	1.313	42.0
Panel C: Median values for groups formed by levels of capital expenditure (MCEV)								
Low	4,537	7.196	22.940	11.460	4.276	82.529	1.244	51.0
Medium	4,691	5.710	24.346	11.100	8.901	102.203	1.198	55.0
High	4,654	5.043	19.908	11.655	16.194	148.939	1.180	53.5
Panel D: Median values for groups formed by levels of firm age (AGE)								
Young	4,071	5.333	8.805	13.390	9.029	53.580	1.427	24.0
Adult	4,773	5.692	22.062	10.990	8.825	93.345	1.214	51.0
Old	4,842	6.226	32.406	10.680	8.937	241.231	1.064	84.0

[a]Based on 13,882 firm-years, except for the *AGE* variable, where only 13,686 firm-years of data are available. *n* refers to the number of firm-years of data available. All median variables are recalculated for each firm-year using the prior five years' data.

[b]*MROA* = median of income before extraordinary items and discontinued operations divided by market value of equity, expressed as a percentage;
MDP = median of common dividends divided by income before extraordinary items and discontinued operations, expressed as a percentage;
MSG = median annual percentage sales growth;
MCEV = median of capital expenditure divided by the market value of equity plus book value of long-term debt, expressed as a percentage;
MVALUE = median market value of equity plus book value of debt;
β = estimate of market model slope coefficient, using prior five years' data;
AGE = age of the firm, computed as the difference between the current year and the year in which the business was originally formed.

based on median values of each life cycle descriptor. Second, we obtain a combined rank by adding the univariate ranks.[11] Third, we resort industries on the combined rank. Table 4 provides the top and bottom 20 industries based on the combined rank, and indicates an improvement in classification when combining

[11]To make the combined rank meaningful, industries are ranked in ascending order of *IMDP*, *IMAGE*, and descending order of *IMSG*. Thus, industries with smaller combined rank are lower dividend payout, higher sales growth, and newer industries.

Table 4

List of industries sorted by median values of *DP*, *SG*, and *AGE*.

SIC code	Industry name	Firm-years[a]	*IMDP*	*IMSG*	*IMAGE*
	Low DP, High SG, and Young industries				
7372	Cmp program & software svcs	47	0.000	16.140	25.000
3679	Electronic components, nec	118	7.100	14.210	32.500
5945	Hobby, toy, and game shops	40	0.000	13.950	46.500
5065	Electronic parts & eq-whsl	102	7.360	12.560	31.000
3674	Semiconductor, related device	83	0.000	10.240	24.000
7011	Hotels, motels, tourist courts	67	9.750	11.580	25.000
1311	Crude petroleum & natural gas	397	9.980	12.630	32.000
3724	Aircraft engine, engine parts	44	7.190	11.110	30.500
8062	Gen med & surgical hospitals	46	21.830	22.460	20.500
4511	Air transportation, certified	161	0.000	13.110	49.000
3825	Elec meas & test instruments	90	11.240	11.790	31.000
8911	Engr, architect, survey svcs	134	18.910	14.060	34.000
5063	Elec apparatus & equip-whsl	45	15.270	12.670	36.000
3585	Air cond, heating, refrig eq	64	0.510	11.560	46.500
4210	Trucking, local, long distance	42	17.790	14.160	45.000
3990	Misc manufacturing industries	43	21.170	11.850	28.000
5331	Variety stores	166	14.170	11.350	42.000
3680	Electronic computing equip	92	10.610	13.460	53.500
5812	Eating places	124	14.770	11.050	40.500
4833	Television broadcasting	75	22.130	13.840	43.000
	High DP, Low SG, and Old industries				
5311	Department stores	142	33.110	9.020	69.000
3861	Photographic equip & suppl	58	42.230	9.690	68.000
2834	Pharmaceutical preparations	258	38.690	9.950	81.000
3530	Constr, mining, matl handle eq	47	30.080	7.650	65.000
2911	Petroleum refining	364	33.080	8.370	65.500
5051	Metals service centers-whsl	42	23.850	5.830	70.500
2211	Brd woven fabric mill, cotton	56	30.540	9.130	89.500
1600	Construction-not bldg constr	77	26.400	7.730	78.000
2030	Can, preserve fruit, vegetable	80	36.520	4.720	53.500
3290	Abrasive, asbestos, misc minrl	61	24.880	5.390	69.000
2040	Grain mill products	55	37.820	9.230	78.000
2080	Beverages	63	35.520	9.500	99.000
2200	Textile mill products	148	24.080	5.830	81.000
1000	Metal mining	53	32.400	5.360	61.000
3630	Household appliances	79	40.740	8.560	71.000
2421	Sawmills, planing mills, gen	40	46.990	7.870	67.500
2000	Food and kindred products	96	33.590	7.910	81.500
2840	Soap, detergent, toilet preps	42	44.610	9.010	140.500
2821	Plastics, resins, elastomers	44	36.550	6.080	77.500
2800	Chemicals & allied prods	152	36.980	7.820	92.000

[a]Firm-years represent the total number of observations available within the four-digit SIC code over the sample period.

information from the three life cycle descriptors. Most of the top 20 industries in the list are typically cited as 'growth' industries. Similarly, a substantial number in the bottom 20 are from industries where new product developments are more rare.

4.3. *Differential response coefficients across life cycle descriptors*

Table 5 provides the results of regressing *CAR* on accounting performance measures with slope dummies for groups formed by median levels of dividend payout, sales growth, capital expenditure, and firm age, respectively.[12] The test results for our differential response coefficient hypotheses are also presented.[13] We initially adopt the univariate classification procedure to investigate the contribution of each variable to the multivariate grouping scheme. In addition, since sales growth is used both in defining an independent variable and as a grouping variable (ΔSG and MSG, respectively), observed monotonicity in the slope coefficient of ΔSG may be attributed to spurious effects of the classification scheme. The univariate results demonstrate that the trend in the slope coefficient of ΔSG is generally consistent with our predictions, even when MSG is not used as a classification variable.

Results in panel A are consistent with our hypotheses. Slope coefficients of ΔCE and ΔSG decrease from the low to high dividend payout group. The slope coefficient of ΔCE, α_2, for the low MDP group is 0.081 (significant at the 0.01 level), whereas those of the other groups are insignificantly different from zero. In addition, α_2 of the low MDP group is significantly greater than those of the medium and high MDP groups at the 0.05 level. The slope coefficient of ΔSG, α_3, is 0.172 for the low MDP group which is more than twice that of 0.074 for the high MDP group. While α_3 is significantly different from zero for all three dividend payout groups, α_3 of the high MDP group is significantly smaller than those of the low and medium MDP groups. Overall, in four of six cases, we reject the null hypothesis.

Using sales growth as a grouping variable fails to support the capital capacity argument even though the results (see panel B) are consistent with the market share argument. There is a monotonic decline in α_3 from the high to the low MSG group, and α_3 of the high MSG group is more than twice that of the low MSG group. In addition, α_3 of the high MSG group is significantly greater than those of the lower MSG groups at the 0.01 level. With respect to the capital capacity argument, while α_2 of the high MSG group is greater than those of the

[12]Our outlier deletion rule is applied here and in all subsequent regression analyses. While having marginal impact on α_1 and α_2, the deletion rule significantly increases the magnitude and statistical significance of α_3 in all regressions. Regression sample size is reduced to 13,653 firm-years after deleting outliers.

[13]We test the significance of the slope coefficients (difference between the slope coefficients) using two-tailed (one-tailed) t-tests.

Table 5

Regression of *CAR* on differenced accounting performance measures with slope dummies for levels of life cycle descriptors.[a, b]

$$CAR = \sum_{i=1}^{3} D_i[\alpha_{0i} + \alpha_{1i}\Delta IBED + \alpha_{2i}\Delta CE + \alpha_{3i}\Delta SG] + \varepsilon$$

	Levels	n	α_0	α_1	α_2	α_3
	Panel A: Slope dummies for levels of median dividend payout (MDP)					
MDP:	Low (L)	4122	− 0.0176 (− 3.099)***	0.4155 (24.215)***	0.0812 (3.481)***	0.1717 (9.919)***
	Medium (M)	4628	− 0.0582 (− 10.928)***	0.5595 (15.677)***	− 0.0227 (− 0.666)	0.1607 (7.276)***
	High (H)	4903	0.0012 (0.240)	0.5785 (13.713)***	− 0.0123 (− 0.260)	0.0741 (3.0·14)***
	L (−) M				0.1039 (2.514)***	0.0110 (0.391)
	L (−) H				0.0935 (1.773)**	0.0976 (3.247)***
	M (−) H				− 0.0104 (− 0.178)	0.0867 (2.622)***
	Panel B: Slope dummies for levels of median sales growth (MSG)					
MSG:	High (H)	4154	− 0.0882 (− 15.497)***	0.6372 (17.130)***	0.0813 (2.775)***	0.1824 (9.094)***
	Medium (M)	4747	− 0.0237 (− 4.541)***	0.4984 (17.561)***	− 0.0121 (− 0.353)	0.0969 (3.950)***
	Low (L)	4752	0.0333 (6.324)***	0.3991 (21.307)***	0.0648 (2.216)**	0.0825 (4.430)***
	H (−) M				0.0933 (2.075)**	0.0856 (2.701)***
	H (−) L				0.0165 (0.398)	0.0999 (3.652)***
	M (−) L				− 0.0769 (− 1.710)	0.0144 (0.467)
	Panel C: Slope dummies for levels of median capital expenditure (MCEV)					
MCEV:	High (H)	4569	− 0.0013 (− 0.242)	0.4226 (19.848)***	0.0305 (1.452)	0.1017 (4.691)***
	Medium (M)	4626	− 0.0268 (− 4.975)***	0.4615 (17.460)***	0.0751 (1.839)*	0.1807 (8.215)***
	Low (L)	4458	− 0.0484 (− 8.747)***	0.5322 (17.516)***	0.1003 (1.654)*	0.1583 (8.429)***

Table 5 (continued)

	Levels	n	α_0	α_1	α_2	α_3
	H (−) M				− 0.0446 (− 0.972)	− 0.0789 (− 2.557)
	H (−) L				− 0.0698 (− 1.088)	− 0.0566 (− 1.972)
	M (−) L				− 0.0252 (− 0.344)	0.0224 (0.775)
	Panel D: Slope dummies for levels of firm age (AGE)					
AGE:	Young (N)	3981	− 0.0369 (− 6.402)***	0.5520 (22.431)***	0.0806 (2.796)***	0.2256 (11.529)***
	Adult (A)	4670	− 0.0233 (− 4.380)***	0.5480 (21.528)***	0.0133 (0.469)	0.0953 (4.663)***
	Old (O)	4815	− 0.0182 (− 3.498)***	0.2877 (10.772)***	0.0500 (1.134)	0.1202 (5.318)***
	N (−) A				0.0673 (1.664)**	0.1303 (4.602)***
	N (−) O				0.0306 (0.580)	0.1054 (3.526)***
	A (−) O				− 0.0367 (− 0.700)	− 0.0249 (− 0.815)

[a]The number in parentheses is the t-statistic. A *** (**/*) designates statistical significance at the 0.01 (0.05/0.10) level. The significance of the slope coefficients (difference between the slope coefficients) is tested using two-tailed (one-tailed) t-tests. The elimination of outliers reduces the sample size to 13,653 firm-years for these tests (13,466 firm-years for the *AGE* subsample). n refers to the number of firm-years of data available for each subsample. Firm-years are assigned to one of the three groups based on their relative cross-sectional ranking on the grouping variable. As in prior tables, median values are based on the prior five years' data.

[b]*MDP* = median of common dividends divided by income before extraordinary items and discontinued operations (*IBED*), expressed as a percentage;
MSG = median annual percentage sales growth;
MCEV = median of capital expenditure divided by the market value of equity plus book value of long-term debt, expressed as a percentage;
AGE = age of the firm, computed as the difference between the current year and the year in which the business was orginally formed;
CAR = sum of abnormal returns from the fourth month of the relevant fiscal year to the third month following the end of the fiscal year;
D_i = zero/one dummy variables proxying for the level of grouping variable;
$\Delta IBED$ = differenced *IBED* divided by prior-year market value of equity (*MVE*);
ΔCE = differenced capital expenditure divided by prior-year *MVE*;
ΔSG = change in percentage sales growth.

other groups, there is no monotonic decline in α_2 from the high to low *MSG* groups. In addition, while α_2 of the high *MSG* group is significantly greater than that of the medium *MSG* group, it is insignificantly different from that of the low *MSG* group. We reject the null hypothesis in three of six cases.

Grouping firms by level of capital expenditure (panel C) provides response coefficients inconsistent with our predictions on market share and capital capacity. There is a monotonic increase in α_2 from the high to the low *MCEV* group, and α_3 does not provide any discernable pattern. *MCEV* is excluded as a life cycle descriptor in subsequent analysis since understanding inter-industry differences in *MCEV* (see section 4.2) requires knowledge of industry-level production functions, which is beyond the scope of this paper.[14]

Panel D of table 5 presents the results of regressing *CAR* on accounting performance measures for *AGE* subsamples. The results are consistent with our hypotheses. While the magnitude of response coefficients for ΔSG and ΔCE does not monotonically decrease from Young to Old firms, they are largest for the Young firms. In terms of hypothesis testing, while α_3 of Young firms is significantly greater than that of both Adult and Old firms, α_2 of Young firms is significantly greater than that of only Adult firms. Overall, the evidence suggests that the mean benefit–cost ratio of increased market share and capital capacity is a function of firm age, consistent with the life cycle theory.

4.4. Differential response coefficients across life cycle stages

Table 6 presents the results of regressing *CAR* on the accounting performance measures for each of the five life cycle groups formed using the composite score.[15] Results are consistent with our hypotheses. There is almost a monotonic decline in the slope coefficients of ΔCE and ΔSG from the Growth to Stagnant stages, and both α_2 and α_3 are insignificant for the Stagnant firms.

The test results of the differential response coefficient hypotheses also are presented in table 6. Five of the six *t*-tests reject the null hypothesis at the 0.05 level. Not only is α_3 of the Growth stage significantly greater than that of both the Mature and Stagnant stages, but α_3 of the Mature stage is also significantly greater than that of the Stagnant stage. While the Growth stage α_2 is greater than that of both the Mature and Stagnant stages, the Mature stage α_2 is insignificantly different from that of the Stagnant stage.

[14]Ex post we rationalize that a firm's total investment in resources would be closely related to life cycle stage. This requires a much broader measure of investment than simple capital expenditures. For example, a service industry invests heavily in human capital, which is not captured by our capital expenditure measure.

[15]Since the score for each variable ranges from one to three, the composite score ranges from three to nine. Firm-years with scores less than or equal to four (greater than or equal to eight) are assigned to the Growth (Stagnant) group. Firm-years with scores five, six, or seven are assigned to the three intermediate groups.

Table 6

Regression of CAR on differenced accounting performance measures with slope dummies for life cycle groups based on MDP, MSG, and AGE.[a,b]

$$CAR = \sum_{i=1}^{5} D_i[\alpha_{0i} + \alpha_{1i}\Delta IBED + \alpha_{2i}\Delta CE + \alpha_{3i}\Delta SG] + \varepsilon$$

Stage	n	α_0	α_1	α_2	α_3
Growth (G)	2420	− 0.0942 (− 12.656)***	0.5627 (16.513)***	0.1200 (3.536)***	0.2078 (8.473)***
Growth/Mature	2454	− 0.0169 (− 2.307)**	0.5637 (19.471)***	0.0342 (0.976)	0.1786 (7.149)***
Mature (M)	2599	− 0.0217 (− 3.066)***	0.4688 (15.484)***	0.0297 (0.766)	0.1365 (4.996)***
Mature/Stagnant	2666	− 0.0158 (− 2.269)**	0.2627 (8.170)***	0.0515 (0.971)	0.1279 (4.600)***
Stagnant (S)	3327	0.0113 (1.809)*	0.5185 (11.760)***	− 0.0413 (− 0.693)	− 0.0022 (− 0.072)
G (−) M				0.0903 (1.753)**	0.0713 (1.943)**
G (−) S				(0.1613 (2.351)***	0.2100 (5.352)***
M (−) S				0.0710 (0.998)	0.1387 (3.379)***

[a]The number in parentheses is the t-statistic. A *** (**/*) designates statistical significance at the 0.01 (0.05/0.10) level. The significance of the slope coefficients (difference between the slope coefficients) is tested using two-tailed (one-tailed) t-tests. Elimination of outliers and firms with missing AGE data reduces the sample size to 13,466 firm-years for these tests. n refers to the number of firm-years of data available for each subsample. Firm-years are assigned to one of the five life cycle groups based on a composite ranking on the grouping variables such that low (high) MDP, high (low) MSG, and young (old) AGE firms are in Growth (Stagnant) groups. As in prior tables, median values are based on the prior five years' data.

[b]MDP = median of common dividends divided by income before extraordinary items and discontinued operations ($IBED$), expressed as a percentage;
MSG = median annual percentage sales growth;
$MCEV$ = median of capital expenditure divided by the market value of equity plus book value of long-term debt, expressed as a percentage;
AGE = age of the firm, computed as the difference between the current year and the year in which the business was originally formed;
CAR = sum of abnormal returns from the fourth month of the relevant fiscal year to the third month following the end of the fiscal year;
D_i = zero/one dummy variables proxying for life cycle stage;
$\Delta IBED$ = differenced $IBED$ divided by prior-year market value of equity (MVE);
ΔCE = differenced capital expenditure divided by prior-year MVE;
ΔSG = change in percentage sales growth.

Table 7

Average slope coefficients from firm-specific regression of *CAR* on differenced accounting performance measures and results of aggregation tests.[a,b]

$$CAR = \alpha_0 + \alpha_1 \Delta IBED + \alpha_2 \Delta CE + \alpha_3 \Delta SG + \varepsilon$$

			ΔIBED	*ΔCE*	*ΔSG*
Stage	Firms	*n*	Average α_1 (*Z*-stat.)	Average α_2 (*Z*-stat.)	Average α_3 (*Z*-stat.)
Growth (G)	236	2351	1.3899 (15.042)***	0.1548 (1.673)*	0.1690 (3.982)***
Growth/Mature	223	2243	1.0772 (14.735)***	0.0066 (0.273)	0.0358 (0.779)
Mature (M)	226	2298	1.1662 (15.322)***	− 0.0645 (− 0.766)	0.0263 (0.560)
Mature/Stagnant	219	2296	1.0197 (14.612)***	− 0.1161 (− 1.431)	− 0.0028 (− 0.095)
Stagnant (S)	248	2580	0.7806 (8.451)***	− 0.2752 (− 2.634)***	− 0.0419 (− 0.945)
G (−) M				0.2193 (1.731)**	0.1427 (2.454)***
G (−) S				0.4300 (3.053)***	0.2109 (3.457)***
M (−) S				0.2107 (1.376)*	0.0682 (1.070)

[a]Average α's are weighted average means of firm-specific α's with weights inversely proportional to coefficient standard error. Computation of the standard normal *Z*-statistic assumes that firm-specific *t*-statistics are independent. A *** (**/*) designates statistical significance at the 0.01 (0.05/0.10) level. The significance of the average slope coefficients (difference between the average slope coefficients) is tested using two-tailed (one-tailed) tests. *n* refers to the number of firm-years of data available. Firm-years are assigned to one of the five life cycle groups based on a composite ranking on the grouping variables such that low (high) *MDP*, high (low) *MSG*, and young (old) *AGE* firms are in Growth (Stagnant) groups. As in prior tables, median values are based on the prior five years' data.

[b]*MDP* = median of common dividends divided by income before extraordinary items and discontinued operations (*IBED*), expressed as a percentage;
MSG = median annual percentage sales growth;
MCEV = median of capital expenditure divided by the market value of equity plus book value of long-term debt, expressed as a percentage;
AGE = age of the firm, computed as the difference between the current year and the year in which the business was originally formed;
CAR = sum of abnormal returns from the fourth month of the relevant fiscal year to the third month following the end of the fiscal year;
ΔIBED = differenced *IBED* divided by prior-year market value of equity (*MVE*);
ΔCE = differenced capital expenditure divided by prior-year *MVE*;
ΔSG = change in percentage sales growth.

4.5. Firm-specific response coefficients as a function of life cycle stage

In the analysis conducted so far, we classify firm-years into life cycle groups, and estimate the response coefficients based on all firm-years within each group. This approach assumes cross-sectional stability in the response coefficients. In the context of earnings, prior empirical evidence suggests systematic differences in response coefficients across firms [see Easton and Zmijewski (1989), Lipe (1986), Kormendi and Lipe (1987), and Collins and Kothari (1989)]. We adopt an alternative approach, estimating firm-specific response coefficients, to control for this possibility.

CAR is regressed on the three differenced performance measures ($\Delta IBED$, ΔCE, and ΔSG) to estimate firm-specific response coefficients. Median values of dividend payout, sales growth, and age for each firm are used to assign firms to portfolios. A modification of Christie's (1990) Z-statistic is used to test for significant differences between the average response coefficients of the groups.[16]

The multivariate results are presented in table 7. The average α_2 and α_3 monotonically decrease from the Growth to the Stagnant groups, and five of the six Z-tests of the differential response coefficients reject the null hypothesis at the 0.10 level. The firm-specific analysis reported in table 7 supports the conclusions from the pooled cross-sectional analysis reported in table 6.

5. Alternative explanations and additional evidence

This section discusses the results of specification checks conducted to investigate the principal alternative explanations of our results.[17]

5.1. Firm size effects

Table 3 indicates that firm size, defined by $MVALUE$, is positively associated with dividend payout and firm age, suggesting that the differential response coefficients could be due to size effects.[18] One would expect that more information is available on the activities of larger firms and that more individuals process and disseminate this information to a broader group of market agents. If prices of larger firms reflect reported earnings earlier, one might expect a similar information effect for capital expenditure and sales growth. In such a case, small

[16]The formulation for the modified Z-statistic is available from the authors.

[17]We briefly describe the additional specification tests performed and the results thereof. The results of the firm size tests (section 5.1) are presented in table 8 to give readers a flavor for the nature and results of the specification tests. All other results are omitted, but available from the authors on request.

[18]Similar effects have been documented by Freeman (1987) and Collins, Kothari, and Rayburn (1987). The impact of firm size is mitigated by the observed positive association between sales growth and firm size.

firms' response coefficients for ΔCE and ΔSG would be higher than those of large firms, and the larger response coefficients for the growth firms would be driven by the smaller growth firms.

To test this argument, we separately regress CAR on differenced performance measures for three size portfolios (defined by $MVALUE$), and report the response coefficients of ΔCE and ΔSG in panel A of table 8 (row 1). The decline in α_2 and α_3 from Small to Large firms is less pronounced than that obtained across life cycle groups. The observed monotonicity across size portfolios is not inconsistent with the life cycle theory since one would expect the growth (stagnant) firms to be smaller (larger) firms.

We run the life cycle regression separately for each size portfolio to separate the confounding effects. The slope coefficients in panel A (rows 2 to 4) indicate that, with one exception, α_2 and α_3 monotonically decline from the Growth to the Stagnant groups for all three size portfolios.[19] The magnitude of the response coefficients for the size portfolios is to some extent driven by the life cycle grouping. For example, while the magnitudes of α_2 and α_3 of the Small firms are driven by the Growth firms, those of the Large firms are driven by the Stagnant firms.

The results of t-tests of differences in response coefficients across stages (the $\Delta\alpha_j$s) are presented in panel B of table 8. While all $\Delta\alpha_2$s have the correct sign, only two differences are significant at the 0.10 level. Market share results are stronger. Eight $\Delta\alpha_3$s have the correct sign, and six are statistically significant at the 0.10 level. Although observed differences in response coefficients are not driven by the size effect, the evidence is consistent with firm size acting as another proxy for life cycle stage. However, since we use proxies (e.g., firm size, dividend payout, sales growth, and firm age) for underlying economic characteristics, our tests cannot categorically distinguish between informational effects and life cycle effects without observing such characteristics.

5.2. *Differential risk effects*

Table 3 indicates that firm risk, defined by beta, has a negative association with dividend payout and firm age, suggesting that the observed monotonicity in response coefficients could be due to risk differences across life cycle groups. Unlike the size effect, no theory links the response coefficients of sales growth and capital expenditure to beta. While the dependent variable (CAR) is risk-adjusted, no such adjustment is made for the independent variables, which could spuriously contribute to differences in the response coefficients. For example, the larger response coefficients of the growth firms could be driven by the high beta firms.

[19]Qualitatively similar results are obtained when market value of equity is used as a proxy for firm size.

Table 8

Regression of CAR on differenced accounting performance measures with slope dummies for life cycle groups by levels of firm size.[a,b]

$$CAR = \sum_{i=1}^{5} D_i[\alpha_{0i} + \alpha_{1i}\Delta IBED + \alpha_{2i}\Delta CE + \alpha_{3i}\Delta SG] + \varepsilon$$

	Small firms			Medium firms			Large firms		
	n	α_2	α_3	n	α_2	α_3	n	α_2	α_3
				Panel A: Slope coefficients[c]					
All firms	4090	0.0885	0.2111	4480	0.0422	0.1128	4896	−0.0091	0.0924
Growth (G)	986	0.1865	0.2439	874	0.0742	0.1728	560	0.1053	0.2097
Mature (M)	930	0.0748	0.1451	825	−0.0102	0.0045	844	0.0656	0.2477
Stagnant (S)	568	0.0166	0.1134	941	−0.0384	−0.0350	1818	−0.0853	−0.0585
				Panel B: Results of t-tests of differential response coefficients[d]					
G (−) M		0.1116	0.0988		0.0844	0.1683		0.0397	−0.0379
		(1.112)	(1.499)*		(1.052)	(2.755)***		(0.427)	(−0.514)
G (−) S		0.1699	0.1306		0.1126	0.2078		0.1906	0.2682
		(1.249)	(1.562)*		(0.801)	(3.144)***		(1.939)**	(3.827)***
M (−) S		0.0582	0.0318		0.0282	0.0395		0.1509	0.3062
		(0.428)	(0.373)		(0.192)	(0.530)		(1.532)*	(5.290)***

[a]Firm-years are assigned to one of the five life cycle groups based on a composite ranking procedure such that low (high) dividend payout, high (low) sales growth, and young (old) firms are in Growth (Stagnant) groups. Firm-years are assigned to one of the three size groups based on their relative cross-sectional ranking on market value of equity plus book value of long-term debt.

[b]CAR = sum of abnormal returns from the fourth month of the relevant fiscal year to the third month following the end of the fiscal year;
D_i = zero/one dummy variables proxying for the life cycle stages;
$\Delta IBED$ = differenced $IBED$ divided by prior-year market value of equity (MVE);
ΔCE = differenced capital expenditure divided by prior-year MVE;
ΔSG = change in percentage sales growth.

[c]Elimination of outliers and firms with missing AGE data reduces the sample size to 13,466 firm-years for the multivariate tests. n refers to the number of firm-years in each size/stage combination.

[d]The number in parentheses is the t-statistic. A *** (**/*) designates statistical significance at the 0.01 (0.05/0.10) level, using one-tailed t-tests.

We conduct the life cycle regression separately for three risk portfolios (defined by level of beta). The results are qualitatively similar for all risk portfolios and indicate no discernible relation between risk and the response coefficients of ΔSG and ΔCE, suggesting that the differential response coefficients are not driven by risk differences across firms.

5.3. Differential information content of prices

Since we assume a random walk (RW) expectations model for our independent variables, it is reasonable to expect measurement error in the independent variables if that assumption is violated. It is well known that measurement error in independent variables yields inconsistent slope estimates [Schmidt (1976, pp. 105–115)]. The error in unexpected performance measures, using a RW assumption, may be related to the information content of prices [see Collins and Kothari (1989)]. If the amount of measurement error is a function of the life cycle descriptors, then differential response coefficients across stages could be driven by the measurement error. Since it is difficult to obtain better estimates of the market's expectations of CE_t and SG_t at the beginning of the CAR cumulation period (e.g., analysts' forecasts), an alternative is to begin the return cumulation period at a point such that CE_{t-1} and SG_{t-1} approximate the market's expectations. This approach, suggested by Collins and Kothari (1989), is adopted here by varying the return window. Specifically, we choose a two-year CAR (denoted $CAR2$) as an alternative dependent variable.[20]

The magnitude and statistical significance of α_2 from the $CAR2$ regression increases substantially for all life cycle groups, suggesting that CE_{t-1} may be a better proxy for CE_t at the beginning of the $CAR2$ cumulation period. Altering the CAR cumulation period changes none of our inferences with respect to the life cycle hypotheses.

5.4. Firm-specific time-series model for performance measures

In addition to changing the cumulation period, we estimate firm-specific time-series models for the performance measures to reduce measurement error in the estimation of unexpected values. Systematic differences in the time-series properties of the independent variables across life cycle stages could explain the monotonic decline in the response coefficients of ΔCE and ΔSG. To mitigate this concern, we estimate univariate AR(1) models (with an intercept) for $IBED_t$, CE_t (both deflated by MVE_{t-1}), and SG_t, and compute residuals from the time-series

[20]The cumulation period of $CAR2$ begins twelve months prior to the beginning of the CAR cumulation period. In measuring $CAR2$, β is estimated separately for each of the two one-year components of $CAR2$.

model.[21] The life cycle regression is replicated with time-series residuals replacing the differenced performance measures as independent variables. The results are consistent with those in table 6.

5.5. *Industry-adjusted independent variables*

The analysis so far conducted employs changes in sales growth and capital expenditure to proxy for changes in market share and capital capacity, respectively, which has obvious limitations. If a firm increases sales growth by 5% (i.e., positive ΔSG) when the industry increases sales growth by 10%, then the firm is losing market share. Similar arguments can be made for capital capacity. To mitigate this problem, we calculate unexpected values for the independent variables using industry medians as our expectation, rather than a random walk.[22] To investigate the relative role of industry-adjusted and differenced performance measures, we include both in a single regression model.

The results of this regression and t-tests of differential response coefficients indicate that differencing (industry adjustment) provides results more consistent with our theory for the sales growth (capital expenditure) variable.[23]

6. Limitations and concluding remarks

We test whether the stock market response to accounting performance measures is a function of the life cycle stage of a firm, hypothesizing that response coefficients of unexpected sales growth and unexpected capital expenditure decrease from the growth to the stagnant portfolios. We test our hypotheses, regressing market model abnormal returns on accounting performance

[21]The choice of an AR(1) model is influenced by the limited time-series observations available. The number of time-series observations ranges from 6 to 17 with a median of 15. An intercept is estimated to capture any potential mean reversions in the variables. The parameter estimates and the residuals are obtained using the same set of time-series observations. These results are available from the authors.

[22]We include all industry-years with at least three firms, which results in a total of 2,177 industry-years (with three to 61 firms) and a median (mean) of five (6.89) firms. However, the median firm in each industry-year is eliminated in the regressions since the unexpected value for that firm is zero by construction.

An alternative is to construct direct measures of market share for the sales and capital capacity variables. The validity of industry definition is much more crucial in computing market share compared to our industry adjustment. While a single outlier (or misclassification) could materially affect the market share numbers, it has no significant effect on the computation of industry medians.

[23]While unexpected change in market share (of sales or capital capacity) is the variable of interest, we use first-differenced and industry-adjusted sales growth/capital expenditures as two proxies. While the first proxy ignores the industry effects, the second fails to control for the expected portion of the change in market share. Therefore, in different settings, either of the two proxies could be a better measure of the unexpected change in market share. This could explain why differencing (industry adjustment) provides results more consistent with our theory for the sales growth (capital expenditure) variable. We thank the referee for this suggestion.

measures (earnings, sales growth, and capital expenditure), using pooled cross-sectional data for the various life cycle groups.

In the univariate analysis, the response coefficients of unexpected sales growth and unexpected capital expenditure are higher for low dividend payout (young/high sales growth) compared to high dividend payout (old/low sales growth) firms. For the multivariate procedure, the results indicate a nearly monotonic decline both in the magnitude and statistical significance of the response coefficients of unexpected sales growth and unexpected capital expenditure from the growth to the stagnant portfolios. Additional analysis suggests that this relation is not driven by a firm size effect, risk differences, differences in the time-series properties of performance measures across life cycle groups, or differential information content of prices with respect to performance measures.

Our inferences depend on the quality of our life cycle proxies and the absence of correlated omitted variables. While the descriptive analysis suggests that our descriptors are reasonable proxies for life cycle stage, they could proxy for other economic phenomena. The sample period (1970–86) might be systematically affected by macro economic factors (business cycles, political climate, international business environment, etc.) unrelated to life cycle stages. Our sample suffers from self-selection bias, since we do not have many start-up firms.

Overall, our results indicate a differential role of accounting performance measures across life cycle stages, and highlight the role of nonearnings data in explaining stock returns.

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